PREP 6, LESSONS 11–12

Power, mains and the National Grid

P = V I and P = I^2 R; energy transferred; mains electricity, safety and the Grid

SCORE / 10

Name _____ Class _____ Due _____

Do Now REMEMBER

From earlier lessons, 1 mark each

1 Describe how the resistance of a thermistor varies with temperature. DN 105 [1]

As the temperature increases, the resistance decreases.

2 Describe how the resistance of an LDR varies with light intensity. DN 106 [1]

As the light intensity increases, the resistance decreases.

3 Finding the I–V characteristic of a component: name the independent and dependent variables. DN 121 [1]

Independent: p.d. across the component; dependent: current through it

Key ideas LESSONS 11–12

Show your working

4 A kettle is connected to the 230 V mains. The current is 8.0 A. Use P = V I

Calculate the power of the kettle. [2]

P = V I = 230 × 8.0 ✓

P = 1840 W ✓

5 The current in a 15 Ω resistor is 2.0 A. Use P = I^2 R

Calculate the power transferred in the resistor. [2]

P = I^2 R = 2.0^2 × 15 ✓

P = 60 W ✓

6 The frequency of the UK mains supply is (50 Hz) | 60 Hz | 230 Hz [1]

7 State the function of the earth wire. [1]

A safety wire that stops the appliance becoming live ✓

8 Explain why step-up transformers are used before electricity is transmitted. [1]

Higher p.d. means a smaller current, so less energy is wasted heating the cables ✓

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